Certifying Safe Operating Regions under Monotone Failure Modes

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Abstract

Many systems are configured by a few continuous knobs, and each setting can fail in more than one way: a language model sampled at low temperature degenerates into repetition and at high temperature into incoherence; a laser powder bed fusion process at low energy leaves unmelted powder and at high energy drills keyhole pores. Practitioners need the *safe operating region*, the settings where every failure mode is rare. Each failure rate can only be estimated from trials at that very setting, so evaluating a grid of configurations costs in proportion to its area. We study *certifying a safe region with a family-wise guarantee when every configuration needs its own trials*, using the one piece of structure these systems share: each failure mode is monotone in known directions. We give (i) the *cone rule*, a closed-testing procedure with betting e-processes that is valid at any stopping time and under any adaptive choice of trials, and under monotonicity reduces to two cumulative sums; (ii) a *search algorithm* for the fixed-confidence problem that stops on its own once the target region is certified; (iii) lower bounds showing that with the ordering the cost is governed by the region's *edge* and without it by its *area*, and an upper bound for the search within logarithmic and multiple-testing factors of the ordered lower bound. On sampling settings of a language model [TBD: LM numbers], on additive manufacturing process windows and on a drug-combination screen, the method needs 1.3–9$\times$ fewer trials than adaptive testing without the ordering and, on 48×48 grids, 19–26$\times$ fewer than a full grid with Learn-then-Test fixed-sequence testing, with no unsafe setting certified in any run, while a Gaussian-process level-set baseline certifies unsafe settings in 7.5–100% of runs unless it certifies almost nothing.

1 Introduction

A team deploying a language model must choose sampling settings (temperature, top-p). Too low, and outputs repeat themselves; too high, and they stop making sense. They want a region of settings where, with 95% confidence, both failure rates are below 10%, so that any setting in it is safe to ship. A setting's failure rates can only be estimated by generating outputs at that setting and judging them, which is expensive at scale. The same pattern appears in process qualification for additive manufacturing (laser power $\times$ scan speed; lack of fusion vs. keyholing), in dose finding for drug combinations (toxicity vs. lack of response), and in operating envelopes of controllers.

What we ask for. A certified region C such that, with probability at least $1 - \alpha$, every setting in C has every failure rate below its tolerance. The guarantee must hold when the experimenter looks at results after every batch and stops whenever they like, and however the next trials are chosen, including by an imperfect simulator or model.

Why existing tools fall short. *Risk control for configurations* (Learn-then-Test [Angelopoulos et al., 2025]; Pareto Testing [Laufer-Goldshtein et al., 2023]; Conformal Language Modeling [Quach et al., 2024]; MultiRisk [Joshi et al., 2025]) evaluates every candidate configuration on a shared calibration set. When evaluating a configuration means fresh trials at it, that costs the full grid; we compare against Learn-then-Test's fixed-sequence testing, which uses monotonicity along one axis, and find it needs 19–26$\times$ more trials than our method at 48×48. *Adaptive multiple testing with bandits* [Jamieson and Jain, 2018, Xu et al., 2021] samples adaptively but treats configurations as unrelated. *Monotone thresholding bandits* [Garivier et al., 2017, Cheshire et al., 2020] use order in one dimension, for one constraint, without anytime validity. *Safe Bayesian optimization and GP level-set estimation* [Sui et al., 2015, Gotovos et al., 2013, Losalka and Scarlett, 2023, 2024] rely on the GP model being right; their certificates fail when it is not (Section 6).

Structure. In all these systems each failure mode is monotone in known directions (repetition falls with temperature, incoherence rises with it). The null hypotheses "setting x is unsafe for mode m" are therefore logically related: if x is unsafe, so is everything riskier than x.

Contributions.

1. *Problem.* Certification of a safe region under per-mode partial orders when each configuration needs its own trials, in a fixed-budget and a fixed-confidence version (Section 2).
2. *Testing.* The cone rule: closed testing with weighted e-values, which under monotonicity is a cumulative sum; valid at every time under arbitrary adaptive sampling (Section 3).
3. *Allocation.* For a fixed budget, trials at the riskiest uncertified settings of a slightly shrunken target (the lower bound's binding constraints). For fixed confidence, a galloping/bisection search along the ordering with a two-sided probe that never stalls near a threshold (Section 4).

4. *Theory.* Lower bounds with and without the ordering (edge vs. area); upper bounds by Wald's identity; the search is within $O(\log n)$ and multiple-testing factors of the ordered lower bound (Section 5).

5. *Experiments.* Sampling settings of a language model, additive manufacturing (a Rosenthal melt-pool model and a synthetic window), and a drug-combination screen (Section 6).

2 Problem

Configurations form a grid $\mathcal{X} = [n_1] \times [n_2]$, $N = n_1 n_2$. There are M failure modes. A trial at x yields $Y \in \{0,1\}^M$, i.i.d. given x (modes may be dependent). The failure rate is $p_m(x) = \mathbb{E}Y_m$; x is *safe* if $p_m(x) < \varepsilon_m$ for all m, and the safe region is W.

Assumption A (monotone failure modes)**.** *For each m there is a known product order $\preceq_m$ on $\mathcal{X}$ such that $x \preceq_m y$ implies $p_m(y) \le p_m(x)$ ("y is at least as safe as x for mode m").*

Protocol. Trials are requested in batches; each batch is chosen from all data so far and any side information (a simulator, a cheaper model, a pilot study) before its outcomes are seen.

Fixed budget. Output certified sets C_t with $\mathbb{P}(C_t \subseteq W\ \forall t) \ge 1-\alpha$, and make C_t large.

Fixed confidence. For a slack $q < 1$ and target $T = \{x : p_m(x) \le q\varepsilon_m\ \forall m\}$, stop at a time τ with a certified set C.

Definition 1. *A procedure is* (α,β)-correct *if for every instance satisfying Assumption A: (validity)* $\mathbb{P}(C \not\subseteq W) \le \alpha$*, and (completeness)* $\mathbb{P}(\tau < \infty,\ T \subseteq C) \ge 1-\beta$.

Settings with rates between $q\varepsilon$ and ε may be certified or not.

3 Testing: the cone rule

Per-configuration evidence. For each (m,x), a betting e-process $K_{m,x}(t) = \prod_{s \le t:\, x_s = x}\big(1 + \lambda_s(\varepsilon_m - Y_{s,m})\big)$, with $\lambda_s \in [0, c/(1-\varepsilon_m)]$, $c < 1$, chosen before trial s. Under $H_{m,x} : p_m(x) \ge \varepsilon_m$ it is a nonnegative supermartingale for the filtration generated by all trials and side information (Lemma 1). We bet with a hedge: half a plug-in bet seeded by side information [Waudby-Smith and Ramdas, 2024], half a mixture of eight fixed bets.

The rule. Fix weights $w_{m,x} \ge 0$ with $\sum w \le 1$ before the trials. Let $\mathcal{C}_m(x) = \{y : y \preceq_m x\}$ be the riskier cone of x (including x). Certify x for mode m once

$$S_m(x,t) = \sum_{y \in \mathcal{C}_m(x)} w_{m,y} K_{m,y}(t) \ \ge\ 1/\alpha, \qquad (1)$$

and certify x once it is certified for every mode. For a product order on a grid, S_m for all x is two cumulative sums.

Theorem 1 (validity)**.** *Under Assumption A, for any allocation and stopping time,*

$$\mathbb{P}\big(\exists\, t, (m,x) :\ p_m(x) \ge \varepsilon_m,\ x \text{ certified for } m \text{ by } t\big) \le \alpha.$$

Proof sketch. The unsafe pairs D are closed toward risk, so a false certification of x gives

$$1/\alpha \le S_m(x,t) \le E(t) := \textstyle\sum_{(m,y)\in D} w_{m,y} K_{m,y}(t),$$

a nonnegative supermartingale starting at most 1; apply Ville's inequality. Full proof in Appendix A.2. □

This is closed testing [Marcus et al., 1976] with the weighted e-value intersection test [Vovk and Wang, 2021]: monotonicity makes the smallest true intersection containing (m,x) a cone, so the closure is a cumulative sum. A certificate at x automatically covers every safer setting, and evidence from riskier settings is pooled.

Weights. Certifying x with weight w costs about $\ln(1/(\alpha w))/\mathrm{KL}(p\|\varepsilon)$ trials, so weights should sit on likely edge points. We use the frequency with which a setting is an edge point of the target under perturbed side information (logit shifts), mixed 50/50 with uniform weights.

4 Allocation

Fixed budget: the edge rule. Estimate the target with slack (all estimated rates below 0.7ε); for each mode, run trials at the riskiest uncertified settings of that estimate, several per batch, cheapest first; then re-check uncertified settings whose lower confidence bound still admits membership. These are the binding constraints of the lower bound (Theorem 2a).

Fixed confidence: search. For each mode and each row (a chain of the order), run a *two-sided probe* at one setting at a time: certify it by the cone rule, or reject "$p_m(x) \le q\varepsilon_m$" with a failure-betting e-process at level $\beta/(MN)$. A certificate spreads to safer settings, a rejection to riskier ones. Start at the edge found in an adjacent row (or the side information's predicted edge), gallop until the edge is bracketed, then bisect. Stop when every row is decided for every mode. Because $q < 1$, at every rate at least one side of the probe has positive drift, so no probe stalls.

5 Theory

Let $L = (1-\beta)\ln(1/\alpha) - \ln 2$.

Theorem 2 (lower bounds). *(a) With the ordering, any (α,β)-correct procedure uses at least $L \cdot \mathrm{LP}(T)$ trials in expectation, where the LP's binding constraints are T's riskiest points for each mode (at most $\min(n_1, n_2)$ per mode), plus, in the fixed-confidence problem, a boundary term over the settings z just outside T of $\min\{L/\mathrm{KL}(p(z)\|\varepsilon),\ L_\beta/\mathrm{KL}(p(z)\|q\varepsilon)\}$. (b) Without the ordering, every setting must be resolved individually, giving a sum over all of $\mathcal{X}$.*

Theorem 3 (oracle upper bound). *Running trials only at a cover of T costs at most $\sum_{cover}(\ln(16/(\alpha w)) + b)/g(p)$ in expectation, where $g(p)$ is the best growth rate of the fixed bets; $\mathrm{KL}(p\|\varepsilon)/g(p) \in [1.00, 1.05]$ for $p \in [0.5\%, 4\%]$ at $\varepsilon = 5\%$.*

Theorem 4 (search). *The search is (α,β)-correct, and its expected cost is at most $\sum_{m,j} P_{m,j}(C^* + B)$, where*

$$P_{m,j} \le 2\lceil \log_2(d_{m,j}+1)\rceil + 2$$

is the number of probes in row j for mode m when the first probe is $d_{m,j}$ positions from the edge, $C^* = \max_p \min(T_c(p), T_r(p)) < \infty$ is the largest expected two-sided probe length, and B is the batch size.

Corollary 1 (edge vs. area). *On an $n \times n$ grid the ordered bounds involve $O(Mn)$ settings and the unordered bound $O(n^2)$; the search is within $O(\log n)$, the ratio $C^*/C(p_{boundary})$, and the multiple-testing factor $\ln(16MN/(\alpha w))/L$ of the ordered lower bound.*

Proofs are in Appendix A.

6 Experiments

Applications. *Language model sampling.* SmolLM2-135M [Allal et al., 2025]; 12 temperatures (0.2–2.0) $\times$ 12 top-p values (0.5–1.0) [Holtzman et al., 2020]; prompts from WikiText-2 [Merity et al., 2017]; *degenerate* = more than 20% of 4-grams repeated [Welleck et al., 2020], *incoherent* = mean token NLL under the unmodified model above 3.5 nats; $\varepsilon = 10\%$ for both. Ground truth from 160 generations per setting (monotone fit); side information is a pilot of 16 separate generations per setting, counted in the budget. *Additive manufacturing.* A Rosenthal melt-pool model [Rosenthal, 1946] of 316L with lack-of-fusion [Tang et al., 2017], keyhole [King et al., 2014] and balling criteria, and a synthetic window with controllable simulator error. *Drug combinations.* Toxicity and non-response surfaces with a synergy term; Bliss independence [Bliss, 1939] as side information.

Table 1: Trials to certify 50% of the safe region (fixed budget, median of 5 runs; full grids: smallest budget reaching 50% on average).

	grid	ours	no order	grid FST	grid Bonf.
synthetic	16^2	7.6k	24.6k	58.1k	77.1k
	32^2	16.6k	116k	256k	339k
	48^2	29.6k	270k	576k	763k
Rosenthal 316L	16^2	5.3k	7.7k	112k	135k
	32^2	19.5k	38.0k	493k	653k
	48^2	42.5k	93.5k	1.11M	1.77M
LM	12^2	[TBD:]	[TBD:]	[TBD:]	[TBD:]

Table 2: Fixed confidence on the synthetic window (good side information; median of 4 runs). The search stopped on its own and certified the whole target in every run.

grid	search	ordered LB	ratio	no ordering
8^2	14.2k	3.5k	4.0	21.9k
16^2	50.3k	13.1k	3.8	100.0k
32^2	165.6k	32.8k	5.1	—
48^2	334.6k	40.4k	8.3	—
LM 12^2	[TBD:]	[TBD:]	[TBD:]	[TBD:]

Baselines. *No ordering*: the same e-processes, weights and allocation, without the cone (each setting on its own). *Full grid, Bonferroni*: equal trials everywhere, exact binomial bounds at level $\alpha/(MN)$. *Full grid, fixed sequence*: equal trials everywhere, fixed-sequence testing along each row from the safest end at level $\alpha/(Mn)$, certificates spread to safer settings; this is how Learn-then-Test uses a monotone risk. *GP level set*: a GP classifier surrogate with a confidence-bound certificate.

Fixed budget. Table 1. The gap to the unordered method grows with resolution (synthetic: 3.2$\times$ at 16^2, 9.1$\times$ at 48^2), as the edge-vs-area bounds predict. The full grids pay for the whole area; fixed-sequence testing helps them by only 20–40%. At 24×24, to certify 70% of the region, ours needs 5.0$\times$ fewer trials than no ordering on the synthetic window, 1.5–1.7$\times$ on Rosenthal 316L and 1.3$\times$ on the drug screen.

Fixed confidence. Table 2. Most of the ratio to the lower bound is the multiple-testing price, $\ln(16MN/(\alpha w))$ against L (about 6$\times$ at 32^2); the search uses about 1.6 probes per lower-bound point. With optimistic side information at 32^2 the search needs 215k trials against 458k without the ordering.

Validity. With every setting exactly at the tolerance (the worst case), the false-certification rate is 0–2.75% across configurations at $\alpha = 5\%$ (400 runs each); no unsafe setting was

certified in any efficiency run.

Model-based baseline. The GP level-set certificate certifies an unsafe setting in 40–100% of runs at confidence parameter 2 and still 7.5–20% at 4, where it certifies at most 69% (drug screen) and typically under 25% of the region; ours: 0% and 54–84%. Language model: [TBD:].

Ablations. The ordering and the allocation rule matter most (no ordering +69 to +213%; greedy allocation +63 to +94%); weights, pooling and bet seeding give smaller gains; a fixed maximum bet fails when relevant rates are near the tolerance.

When monotonicity fails. A pocket of unsafe settings inside the region is certified by spreading in 75–92% of runs; an audit re-testing every certified setting (about 1.2× the budget) catches gross violations (96–100%) but rarely mild ones (4–9%).

7 Related work

Anytime-valid inference and e-values: Ville [1939], Shafer [2021], Ramdas et al. [2023], Vovk and Wang [2021], Wang and Ramdas [2022], Waudby-Smith and Ramdas [2024]. *Multiple testing with structure:* closed testing [Marcus et al., 1976], sequential rejection [Goeman and Solari, 2010], graphical procedures [Bretz et al., 2009], closure for monotone bi-factorial dose designs [Hellmich and Lehmacher, 2005]. *Adaptive testing and bandits:* Jamieson and Jain [2018], Xu et al. [2021]; thresholding bandits [Locatelli et al., 2016] and their monotone versions [Garivier et al., 2017, Cheshire et al., 2020, 2021]; lower bounds [Garivier and Kaufmann, 2016, Kaufmann et al., 2016, Garivier et al., 2019]; multi-fidelity best-arm identification [Poiani et al., 2022, 2024]. *Risk control for configurations and LLM evaluation:* Angelopoulos et al. [2025], Laufer-Goldshtein et al. [2023], Quach et al. [2024], Joshi et al. [2025]; anytime-valid evaluation of a single system [Zhou et al., 2026b,a]. *Predictions and simulators inside tests:* prediction-powered inference and e-values [Angelopoulos et al., 2023, Csillag et al., 2025], active inference [Zrnic and Candès, 2024], simulator-guided betting for one robot's mean performance [Chen et al., 2026, Chen and Weng, 2026]; we use side information only for allocation and weights, over a whole region. *Safe exploration and level sets:* Sui et al. [2015], Losalka and Scarlett [2023, 2024], Gotovos et al. [2013], Mason et al. [2022]. *Applications:* adaptive experimental design for AM process parameters [BEAM authors (Washington State University; incl. J. R. Doppa), 2026]; dose finding for combinations [Wages et al., 2011].

8 Limitations

Assumption A is load-bearing (Section 6); thin regions gain less at coarse resolution; the multiple-testing factor dominates the gap to the lower bound; the language-model study uses a small model and automatic judges; the manufacturing studies are simulated with placeholder calibrations.

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A Proofs

A.1 Setup

Trial $t = 1, 2, \ldots$ is run at x_t and yields $Y_t \in \{0,1\}^M$. Given the past and $x_t = x$, $Y_t \sim P_x$, independently of the past. Let $\mathcal{F}_t = \sigma(x_1, Y_1, \ldots, x_t, Y_t, U)$, where U is independent side information (simulator output, pilot data, internal randomization). The allocation x_t is $\mathcal{F}_{t-1}$-measurable: trials in a batch are chosen before any of them is observed, and processing them one at a time in a fixed order keeps this property.

Lemma 1. *If $p_m(x) \geq \varepsilon_m$ then $K_{m,x}$ is a nonnegative supermartingale with respect to $(\mathcal{F}_t)$. If $p_m(x) \leq q\varepsilon_m$, the rejection process $R_{m,x}(t) = \prod_{s \leq t : x_s = x}(1 + \mu_s(Y_{s,m} - q\varepsilon_m))$, $\mu_s \in [0, 1/(q\varepsilon_m))$ predictable, is a nonnegative supermartingale.*

Proof. Nonnegativity: $1+\lambda(\varepsilon-1) \geq 1-c > 0$. On $\{x_t = x\}$, $\mathbb{E}[K(t) \mid \mathcal{F}_{t-1}] = K(t-1)(1+\lambda_t(\varepsilon - p_m(x))) \leq K(t-1)$, since λ_t and x_t are $\mathcal{F}_{t-1}$-measurable and the conditional law of Y_t given $\mathcal{F}_{t-1}$ and $x_t = x$ is P_x. On $\{x_t \neq x\}$, $K(t) = K(t-1)$. The rejection process is identical with the roles of failures and successes exchanged. A fixed-weight average of such processes (our hedge) is again a nonnegative supermartingale. □

A.2 Validity of the cone rule

Proof of Theorem 1. Let $D = \{(m, y) : p_m(y) \geq \varepsilon_m\}$. Under Assumption A, if $(m, x) \in D$ and $y \preceq_m x$ then $p_m(y) \geq p_m(x) \geq \varepsilon_m$, so $\{m\} \times \mathcal{C}_m(x) \subseteq D$. Let $E(t) = \sum_{(m,y) \in D} w_{m,y} K_{m,y}(t)$, a nonnegative supermartingale by Lemma 1 with $E(0) \leq 1$. A false certification of $(m, x) \in D$ at t means $1/\alpha \leq S_m(x, t) \leq E(t)$. By Ville's inequality $\mathbb{P}(\sup_t E(t) \geq 1/\alpha) \leq \alpha$. □

Remarks: nothing about the allocation is used beyond predictability; certification is sticky, and the bound covers "ever certified"; with uniform weights $S_m(x) \geq \max_{y \in \mathcal{C}_m(x)} wK_y$, so the cone rule certifies everything Bonferroni with spreading certifies.

A.3 Correctness of the search

Proof of Theorem 4, correctness. Validity is Theorem 1. Completeness: let G be the event that no (m, x) with $p_m(x) \leq q\varepsilon_m$ is rejected directly. By Lemma 1, Ville and a union bound over MN pairs, $\mathbb{P}(G^c) \leq \beta$. On G no $x \in T$ is rejected or excluded: a spread rejection of x for m comes from a direct rejection at some y with $x \preceq_m y$, but then $p_m(y) \leq p_m(x) \leq q\varepsilon_m$, contradicting G; exclusion through another mode is the same argument for that mode. Termination: every probe ends almost surely (if $p < \varepsilon$ the certification statistic has positive expected log-growth under the hedge's best fixed bet; if $p > q\varepsilon$ the rejection process does; since $q < 1$ every p is covered), each probe decides at least one configuration, and $\mathcal{X}$ is finite. So on G, $\tau < \infty$ and $T \subseteq C$. □

A.4 Expected passage times

Lemma 2 (Wald)**.** *Let $Z_k = \sum_{i \leq k} \xi_i$ with ξ_i i.i.d., bounded, $\mathbb{E}\xi = g > 0$, and $\tau = \min\{k : Z_k \geq c\}$. Then $\mathbb{E}\tau \leq (c + b)/g$, $b = \operatorname{ess\,sup} \xi$.*

Proof. $\mathbb{E}\tau < \infty$ for positive drift with bounded increments; by Wald's identity $\mathbb{E}Z_\tau = g\mathbb{E}\tau$, and $Z_\tau \leq c + b$. □

Applied to the log-wealth of the best fixed bet in the hedge (held with weight $1/16$) and $S_m(x) \geq w_{m,x} K_{m,x}$: a setting with rate $p < \varepsilon$ sampled exclusively is certified within an expected $T_c(p) = (\ln(16/(\alpha w)) + \ln(1 + \lambda^*\varepsilon))/g_c(p)$ trials, where $g_c(p) = \max_j \mathbb{E}\ln(1 + \lambda_j(\varepsilon - Y))$. Likewise rejection at $p > q\varepsilon$ within $T_r(p) = (\ln(5MN/\beta) + b')/g_r(p)$. This gives Theorem 3 by summing over the cover (plus B per cover point when certification is checked once per batch).

Proof sketch of Theorem 4, cost. Galloping brackets the edge in $\lceil \log_2(d + 1) \rceil + 1$ probes and bisection closes the bracket in at most as many; probes elsewhere only decide configurations earlier. A probe ends at the first of two passage times, so its expected length is at most $\min(T_c, T_r) \leq C^*$. Overshoot is at most B per probe. Probes for different modes at one configuration share trials, which only reduces cost. □

A.5 Lower bounds

We use the transportation lemma [Kaufmann et al., 2016, Lemma 1] in the general form of Garivier et al. [2019]: for instances ν, ν', a stopping time τ and $E \in \mathcal{F}_\tau$, $\sum_x \mathbb{E}_\nu[N_x(\tau)]\mathrm{KL}(\nu_x, \nu'_x) \geq \mathrm{kl}(\mathbb{P}_\nu(E), \mathbb{P}_{\nu'}(E))$, and $\mathrm{kl}(a, b) \geq a\ln(1/b) - \ln 2$. We restrict to instances with independent modes (a lower bound on a subclass is a lower bound).

Proof of Theorem 2. (a) For each m and $z \in T$, let ν' set $p'_m(y) = \max(p_m(y), \varepsilon_m + \eta)$ for all $y \preceq_m z$ and leave everything else unchanged. ν' satisfies Assumption A: raising the rates on a riskier-closed set to at least a common value keeps every chain monotone. In ν', z is unsafe, so a valid procedure certifies T with probability at most α; under ν it does so with probability at least $1 - \beta$. Letting $\eta \to 0$, $\sum_{y \preceq_m z,\ p_m(y) < \varepsilon_m} \mathbb{E}[N_y]\mathrm{KL}(p_m(y) \| \varepsilon_m) \geq L$. Minimizing the total over these constraints gives $L \cdot \mathrm{LP}(T)$; only the $\preceq_m$-minimal points of T bind, and an antichain of a product order on a grid has at most one point per row and column.

Boundary term: for z just outside T for mode m (every setting safer than z for m is in T for m, and z is in T for the other modes), lower $p_m(z)$ to $q\varepsilon_m - \eta$. This keeps Assumption A, because every setting safer than z has $p_m \leq q\varepsilon_m$. In the alternative $z \in T$ must be certified with probability at least $1 - \beta$, so the procedure either certifies z in the real instance (cost $\geq L/\mathrm{KL}(p_m(z) \| \varepsilon_m)$, infinite if $p_m(z) \geq \varepsilon_m$) or distinguishes the two instances at z (cost $\geq L_\beta/\mathrm{KL}(p_m(z) \| q\varepsilon_m)$, $L_\beta = (1 - \beta)\ln(1/\beta) - \ln 2$).

(b) Without Assumption A the alternatives may change a single configuration, so every configuration contributes its own term. □

Points to check. (1) Lemma 1 with batched allocation; (2) the closed-testing interpretation of the cone rule; (3) that no target configuration is rejected by spreading or exclusion on G; (4) the exact form of the transportation lemma for procedures that may not stop under the alternative; (5) that the alternatives keep Assumption A, and the LP sharing across modes; (6) that the certification and boundary terms add; (7) the probe count with spreading from other rows.